

Week 5

Keel Bone Chicken Project

06/22/26



College of Engineering

This Week

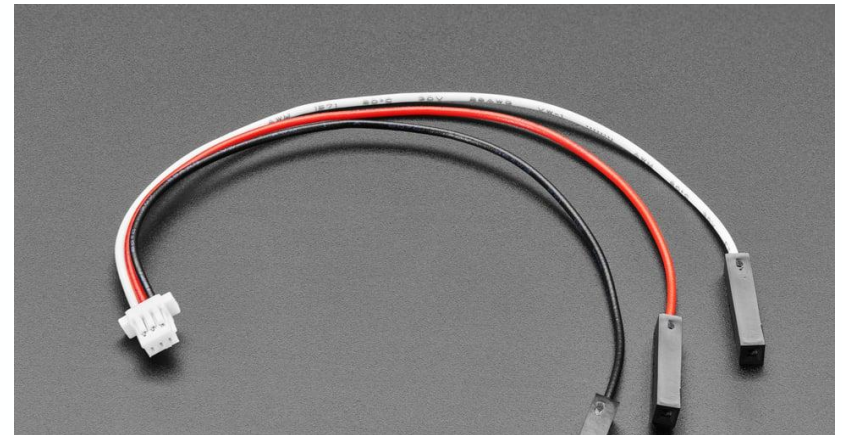
Weekly Progress

- **Hardware Updates:**
 - Agriculture Supplies
- **Testing Updates:**
 - Access to Professor Wu's lab
- **Database Updates:**
 - Database Development still in Progress
- **PCB Updates:**
 - New Routing done and waiting on validation from DL-Engineering
- **Next Week**
 - Goals for next week

Hardware Supplies

Agriculture Supplies

- Supplies requested from Agriculture department
 - 4 Micro USB connectors
 - 3 Connectors (JST to single female dupont)
- 3 Connectors:
- 4 Micro USB connectors:
- Should we purchase them or allow the Agriculture department to purchase them?



Testing Updates

Access to Professor Wu's lab

- Pedro Barbosa (PhD candidate)
 - Able to let me in to allow me to do the testing
- See if any further access or authorization is required
- Testing specifically for piezoelectric devices

Database Updates

Database Development still in Progress

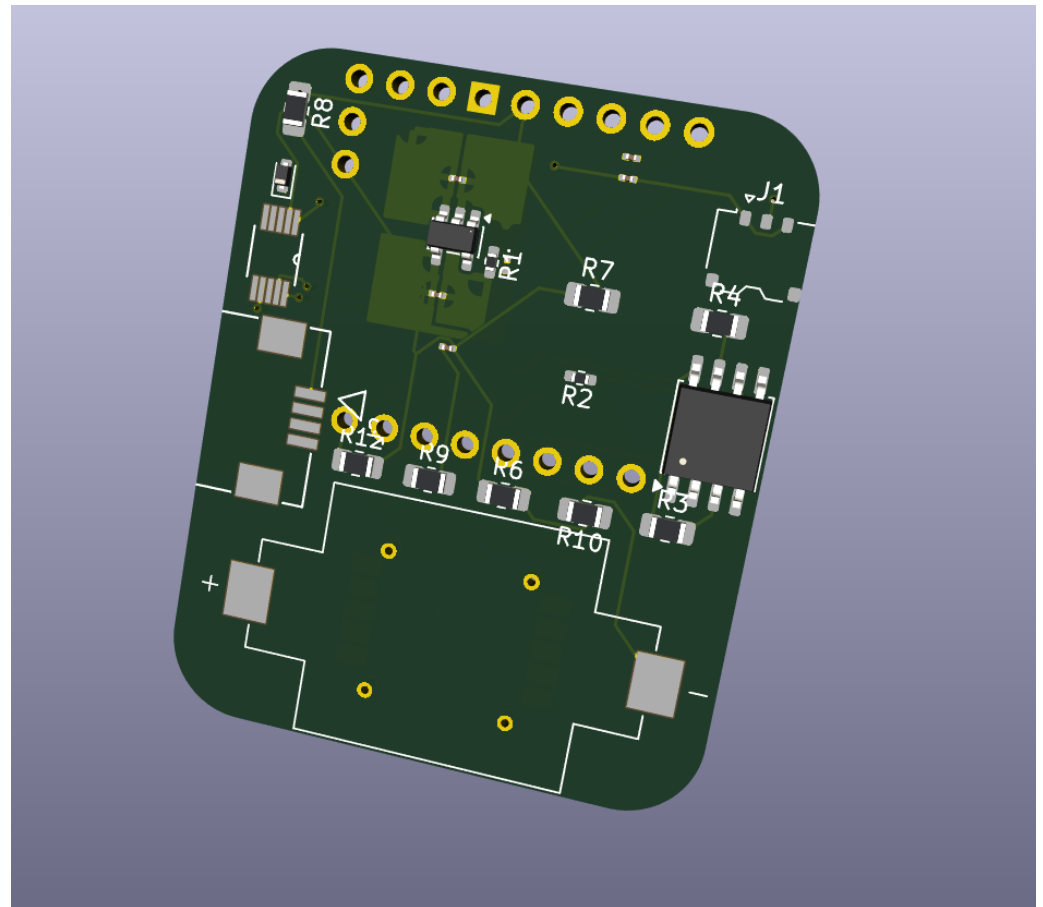
- Testing backend requests
- Finding a more convenient way to extract certain periods of data
 - Was initially custom (0-60 minutes)
 - Want to make intervals for easier exporting
 - 10 min
 - 60 min
 - 3 hours
 - Etc.
 - Find a good match which doesn't overload the backend
- Information about tech stack:

Component	Technology	Critical Function
Runtime	Node.js	Asynchronous event loop for high-frequency data handling.
Web Framework	Express.js	RESTful API endpoints for sensor payload ingestion.
ODM	Mongoose	Schema definition and validation for sensor data objects.
Middleware	CORS / Dotenv	Secure cross-origin requests and environment management.

PCB Updates

New Routing done and waiting on validation from DL-Engineering

- Finalized routing done and waiting on DL-Engineering
- Need to validate with Powercast to ensure device will work on board
- Seen below:



Next Week

Goals for next week

- Coordinate with Pedro at Professor Wu's lab
- Continue Database Development Efforts
- Make final changes to PCB

Thank You

Purdue Polytechnic Institute



College of Engineering